

Assignment 1

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Semester : 7th

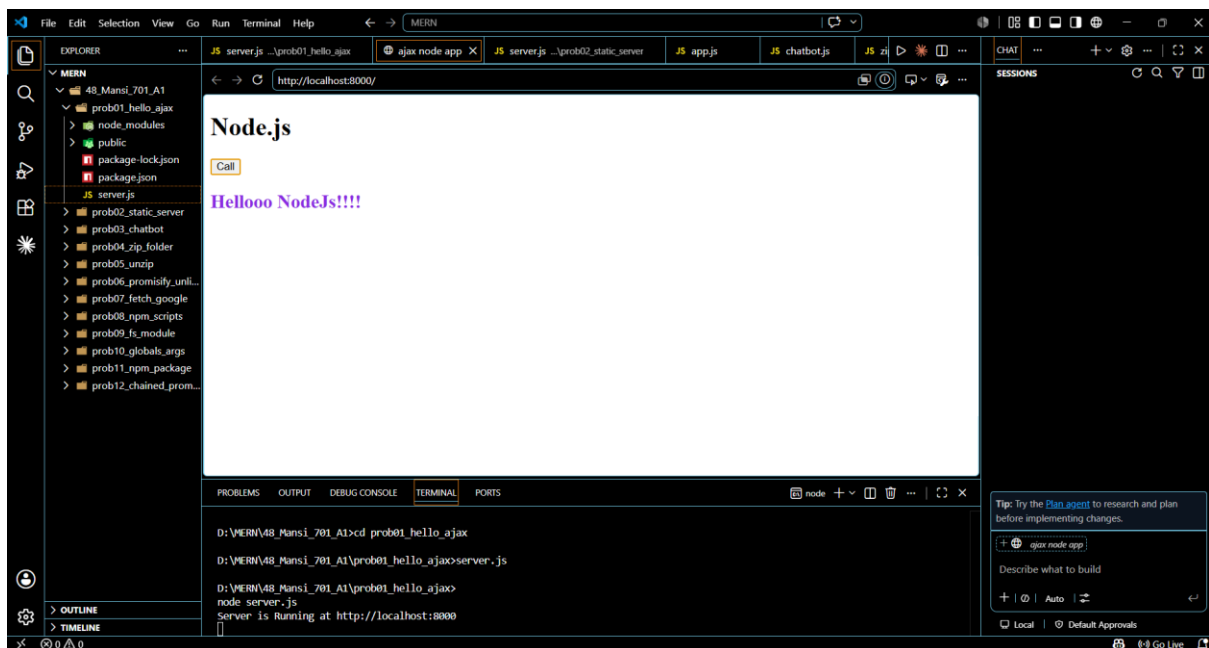
Division : A

Roll Number : 48

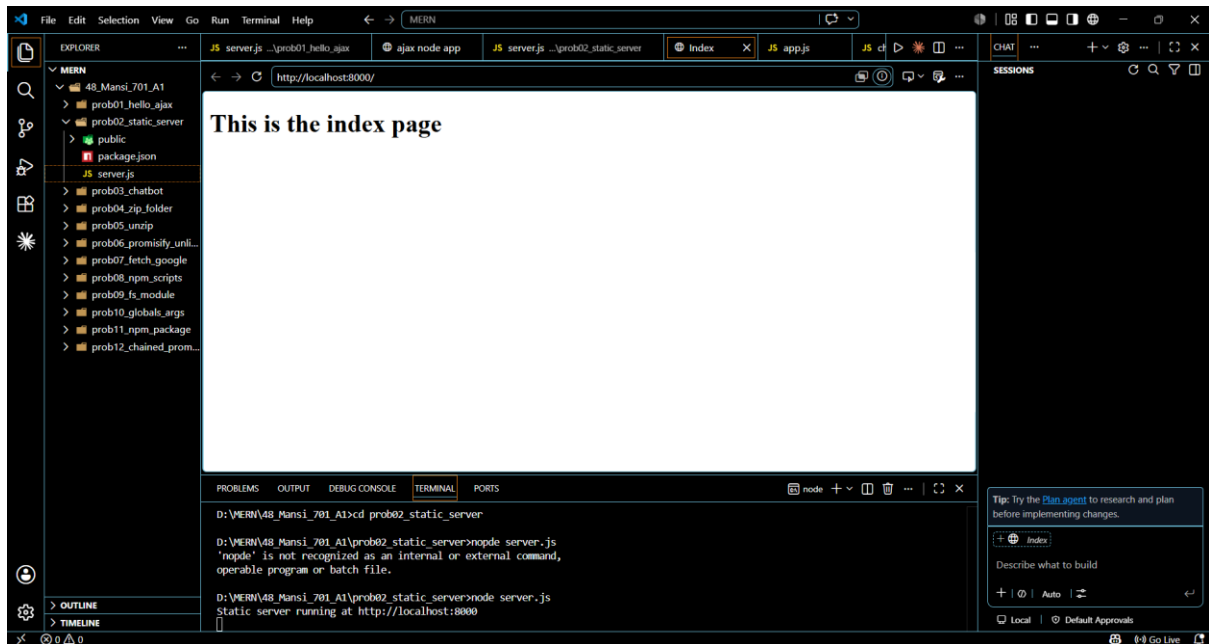
Subject : Application Development using full Stack

Date : 15 July ,2026

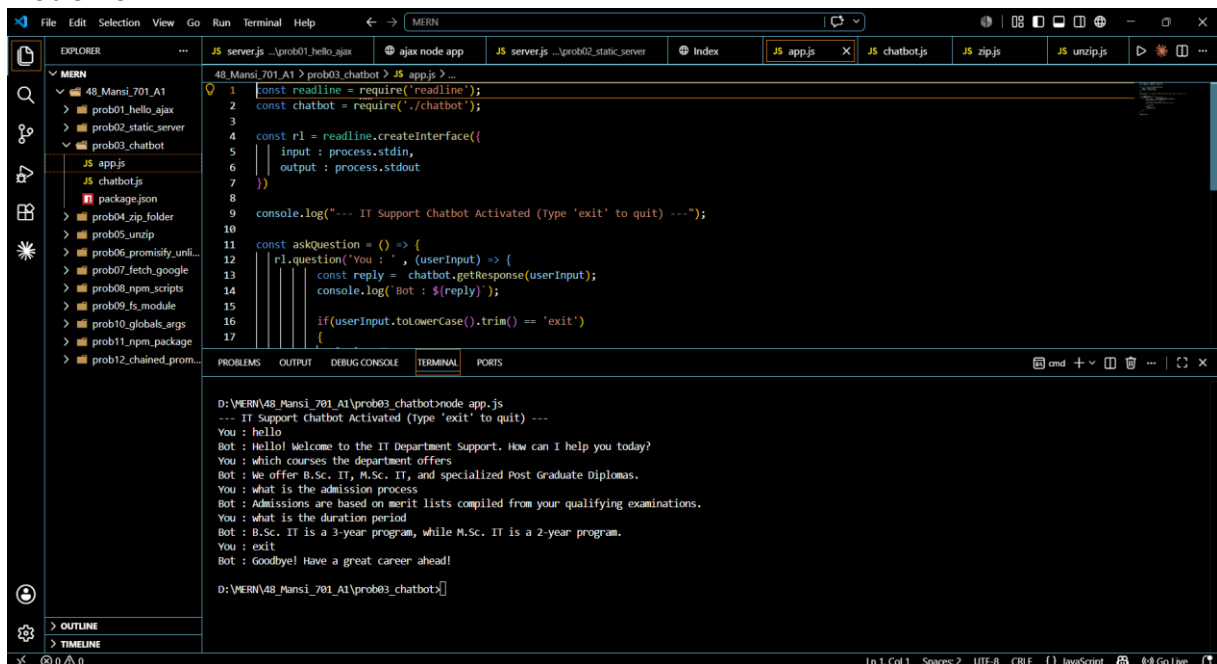
Problem 1 :



Problem 2:



Problem 3



Problem 4 :

```
48_Mansi_701_A1 > prob04_zip_folder > JS zip.js > ...
1 const fs = require("fs");
2 const { ZipArchive } = require("archiver");
3
4 const output = fs.createWriteStream("MyFolder.zip");
5 const archive = new ZipArchive({
6   zlib: { level: 9 }
7 });
8
9 archive.on("warning", (err) => {
10   if (err.code !== "ENOENT") {
11     throw err;
12   }
13 });
14
15 archive.on("error", (err) => {
16   throw err;
17 });
```

Bot : We offer B.Sc. IT, M.Sc. IT, and specialized Post Graduate Diplomas.
You : what is the admission process
Bot : Admissions are based on merit lists compiled from your qualifying examinations.
You : what is the duration period
Bot : B.Sc. IT is a 3-year program, while M.Sc. IT is a 2-year program.
You : exit
Bot : Goodbye! Have a great career ahead!

D:\MERN\48_Mansi_701_A1\prob04_chatbot>cd..
D:\MERN\48_Mansi_701_A1>cd prob04_zip_folder
D:\MERN\48_Mansi_701_A1\prob04_zip_folder>node zip.js
ZIP created successfully!
D:\MERN\48_Mansi_701_A1\prob04_zip_folder>

Problem 5:

```
48_Mansi_701_A1 > prob05_unzip > JS unzip.js > ...
1 const fs = require("fs");
2 const unzipper = require("unzipper");
3 const path = require("path");
4
5 const zipFilePath = path.join(__dirname, "MyFolder.zip");
6 const outputDir = path.join(__dirname, "MyFolder");
7
8 if (!fs.existsSync(zipFilePath)) {
9   console.log("Please copy a valid 'output.zip' into this folder first to test extraction.");
10   process.exit(0);
11 }
12
13 fs.createReadStream(zipFilePath)
14   .pipe(unzipper.Extract({ path: outputDir }))
15   .on("close", () => {
16     console.log("ZIP extracted successfully!");
17   });
```

D:\MERN\48_Mansi_701_A1\prob04_zip_folder>cd..
D:\MERN\48_Mansi_701_A1>cd prob05_unzip
D:\MERN\48_Mansi_701_A1\prob05_unzip>node unzip.js
ZIP extracted successfully!
D:\MERN\48_Mansi_701_A1\prob05_unzip>

Problem 6:

```
48_Mansi_701_A1 > prob06_promisify_unlink > JS promisify.js > ...
1 const fs = require("fs");
2 const path = require("path");
3
4 const targetFile = path.join(__dirname, "temp.txt");
5
6 fs.writeFileSync(targetFile, "Temporary data to be deleted");
7
8 const deleteFileAsync = () => {
9   return new Promise((resolve, reject) => {
10     fs.unlink(targetFile, (err) => {
11       if (err) {
12         reject(err);
13       } else {
14         resolve("file deleted successfully via custome promise");
15       }
16     });
17   });
18 }
19
20 deleteFileAsync()
21   .then((message) => {
22     console.log(message)
23   })
24   .catch((err) => {
25     console.error("Error deleting file:", err)
26   })
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\V\ERN\48_Mansi_701_A1\prob05_unzip> node unzip.js
ZIP extracted successfully!

D:\V\ERN\48_Mansi_701_A1\prob05_unzip>cd..

D:\V\ERN\48_Mansi_701_A1>cd prob06_promisify_unlink

D:\V\ERN\48_Mansi_701_A1\prob06_promisify_unlink>node promisify.js
file deleted successfully via custome promise

D:\V\ERN\48_Mansi_701_A1\prob06_promisify_unlink>

Problem 7 :

static_server Index JS app.js JS chatbot.js JS zip.js JS unzip.js JS promisify.js JS server.js .../prob07_fetch_google Google

http://localhost:8000/google

Gmail Images Sign in

Google

Advanced search

Google Search I'm Feeling Lucky

Google offered in: हिन्दी বাংলা తెలుగు मराठी தமிழ் ગુજરાતી ಕನ್ನಡ മലയാളം ਪੰਜਾਬੀ

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\V\ERN\48_Mansi_701_A1>cd prob06_promisify_unlink

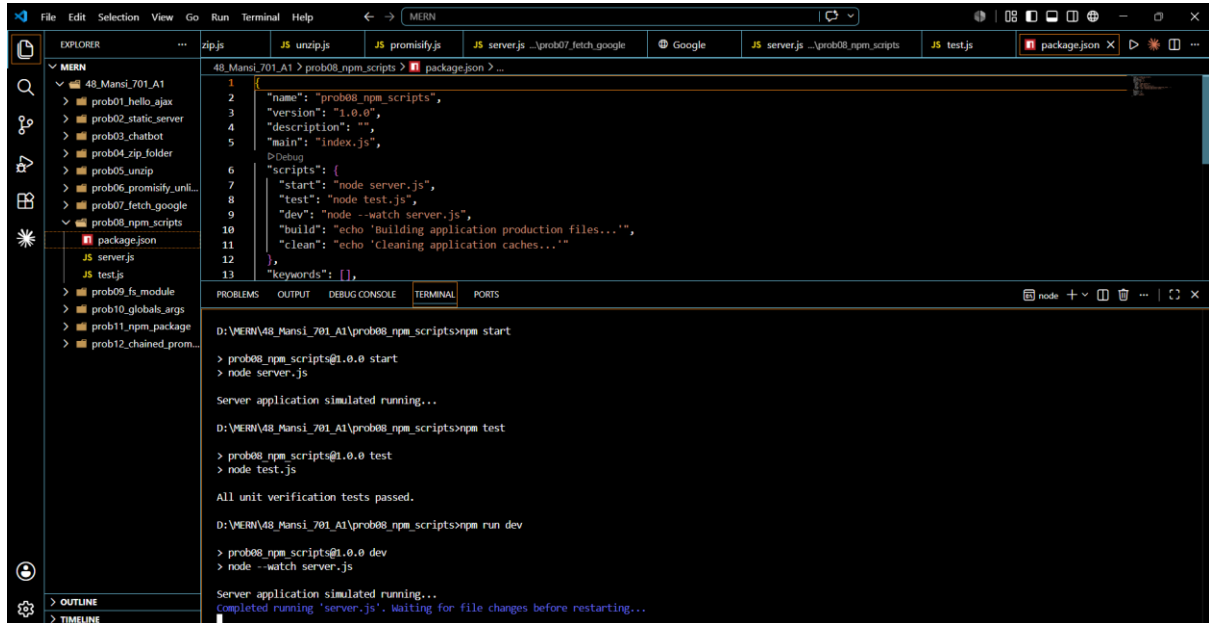
D:\V\ERN\48_Mansi_701_A1\prob06_promisify_unlink>node promisify.js
file deleted successfully via custome promise

D:\V\ERN\48_Mansi_701_A1\prob06_promisify_unlink>cd..

D:\V\ERN\48_Mansi_701_A1>cd prob07_fetch_google

D:\V\ERN\48_Mansi_701_A1\prob07_fetch_google>node server.js
Server handling fetch runs at http://localhost:8000/google

Problem 8:



The screenshot shows the VS Code interface with the Explorer view on the left displaying a project structure. The main editor shows the `package.json` file for the `prob08_npm_scripts` project. The terminal at the bottom shows the execution of `npm start`, `npm test`, and `npm run dev`.

```
48_Mansi_701_A1 > prob08_npm_scripts > package.json > ...
1 {
2   "name": "prob08_npm_scripts",
3   "version": "1.0.0",
4   "description": "",
5   "main": "index.js",
6   "scripts": {
7     "start": "node server.js",
8     "test": "node test.js",
9     "dev": "node --watch server.js",
10    "build": "echo 'Building application production files...'",
11    "clean": "echo 'Cleaning application caches...'"
12  },
13  "keywords": [],
14}
```

Terminal Output:

```
D:\MERN\48_Mansi_701_A1\prob08_npm_scripts>npm start
> prob08_npm_scripts@1.0.0 start
> node server.js

Server application simulated running...

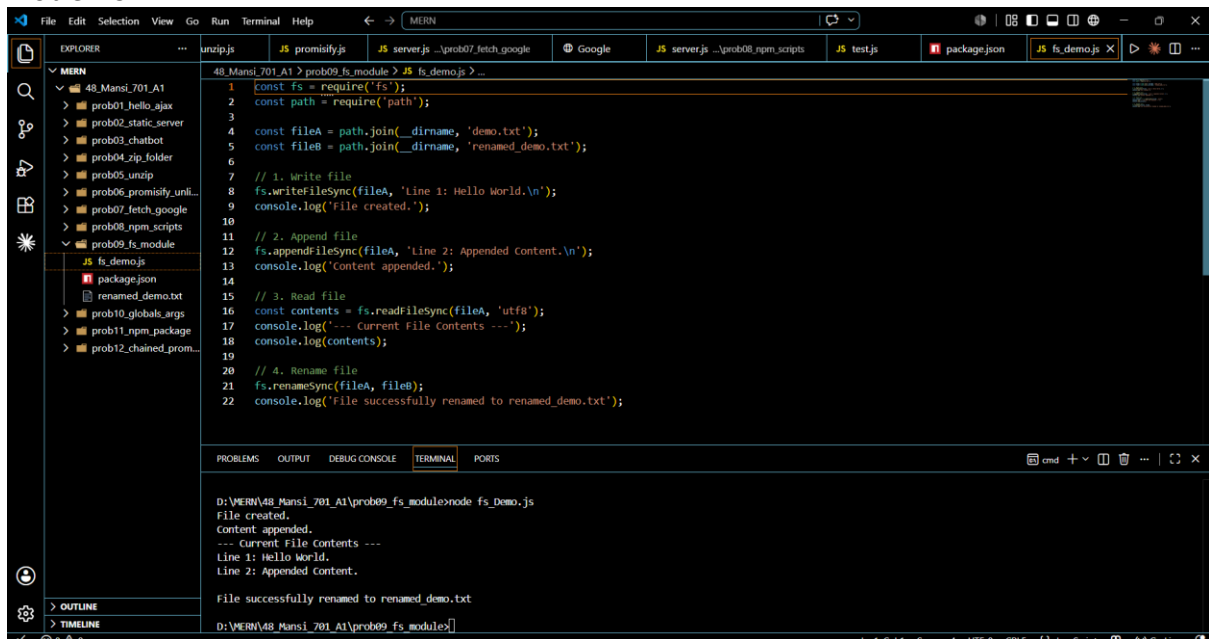
D:\MERN\48_Mansi_701_A1\prob08_npm_scripts>npm test
> prob08_npm_scripts@1.0.0 test
> node test.js

All unit verification tests passed.

D:\MERN\48_Mansi_701_A1\prob08_npm_scripts>npm run dev
> prob08_npm_scripts@1.0.0 dev
> node --watch server.js

Server application simulated running...
Completed running 'server.js'. Waiting for file changes before restarting...
```

Problem 9:



The screenshot shows the VS Code interface with the Explorer view on the left displaying a project structure. The main editor shows the `fs_demo.js` file for the `prob09_fs_module` project. The terminal at the bottom shows the execution of `node fs_demo.js`.

```
48_Mansi_701_A1 > prob09_fs_module > fs_demo.js > ...
1 const fs = require('fs');
2 const path = require('path');
3
4 const fileA = path.join(__dirname, 'demo.txt');
5 const fileB = path.join(__dirname, 'renamed_demo.txt');
6
7 // 1. Write file
8 fs.writeFileSync(fileA, 'Line 1: Hello World.\n');
9 console.log('File created.');
```

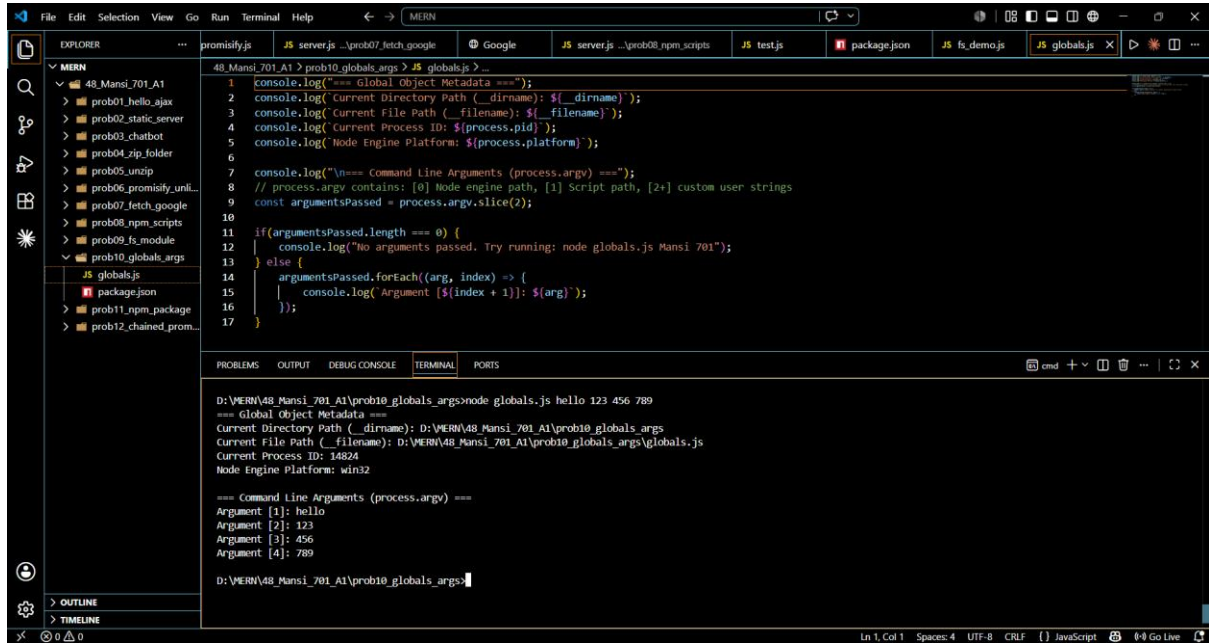
Terminal Output:

```
D:\MERN\48_Mansi_701_A1\prob09_fs_module>node fs_demo.js
File created.
Content appended.
--- Current File Contents ---
Line 1: Hello World.
Line 2: Appended Content.

File successfully renamed to renamed_demo.txt

D:\MERN\48_Mansi_701_A1\prob09_fs_module>
```

Problem 10:



```
48_Mansi_701_A1 > prob10_globals_args > JS globals.js > ...
1 console.log("=== Global Object Metadata ===");
2 console.log("Current Directory Path (._dirname): ${_dirname}");
3 console.log("Current File Path (._filename): ${_filename}");
4 console.log("Current Process ID: ${process.pid}");
5 console.log("Node Engine Platform: ${process.platform}");
6
7 console.log("\n=== Command Line Arguments (process.argv) ===");
8 // process.argv contains: [0] Node engine path, [1] Script path, [2+] custom user strings
9 const argumentsPassed = process.argv.slice(2);
10
11 if(argumentsPassed.length === 0) {
12 | console.log("No arguments passed. Try running: node globals.js Mansi 701");
13 } else {
14 | argumentsPassed.forEach((arg, index) => {
15 | | console.log("Argument ${index + 1}): ${arg}");
16 | });
17 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\VERV\48_Mansi_701_A1\prob10_globals_args>node globals.js hello 123 456 789

=== Global Object Metadata ===

Current Directory Path (._dirname): D:\VERV\48_Mansi_701_A1\prob10_globals_args

Current File Path (._filename): D:\VERV\48_Mansi_701_A1\prob10_globals_args\globals.js

Current Process ID: 14824

Node Engine Platform: win32

=== Command Line Arguments (process.argv) ===

Argument [1]: hello

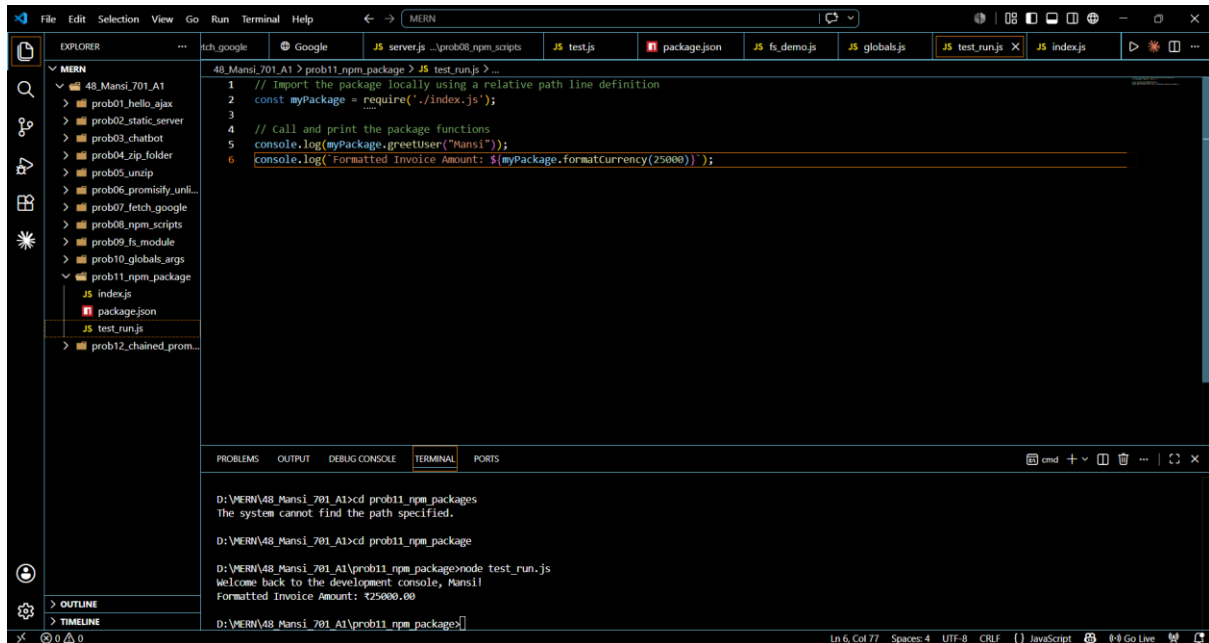
Argument [2]: 123

Argument [3]: 456

Argument [4]: 789

D:\VERV\48_Mansi_701_A1\prob10_globals_args>

Problem 11:



```
48_Mansi_701_A1 > prob11_npm_package > JS test_run.js > ...
1 // Import the package locally using a relative path line definition
2 const myPackage = require('./index.js');
3
4 // Call and print the package functions
5 console.log(myPackage.greetUser("Mansi"));
6 console.log("Formatted Invoice Amount: ${myPackage.formatCurrency(25000)}");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\VERV\48_Mansi_701_A1>cd prob11_npm_packages

The system cannot find the path specified.

D:\VERV\48_Mansi_701_A1>cd prob11_npm_package

D:\VERV\48_Mansi_701_A1\prob11_npm_package>node test_run.js

Welcome back to the development console, Mansi!

Formatted Invoice Amount: ₹25000.00

D:\VERV\48_Mansi_701_A1\prob11_npm_package>

Problem 12

The screenshot shows a VS Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'MERN' with a folder '48_Mansi_701_A1' containing several subfolders and files. The file 'chain.js' is selected. The terminal shows the execution of the 'chain.js' file, which contains three functions: 'verifyUser', 'fetchUserPermissions', and 'generateSessionToken'. The output shows the execution of these functions in sequence, with console logs for each step.

```
48_Mansi_701_A1 > prob12_chained_promises > JS chain.js > verifyUser
1 function verifyUser(userId) {
2   return new Promise((resolve) => {
3     setTimeout(() => {
4       console.log("Task 1 Completed: User credentials validated.");
5       resolve({ id: userId, name: "Mansi Patel" });
6     }, 1000);
7   });
8 }
9
10 function fetchUserPermissions(user) {
11   return new Promise((resolve) => {
12     setTimeout(() => {
13       console.log("Task 2 Completed: Permissions retrieved for ${user.name}.");
14       resolve({ ...user, role: "Administrator", access: ["Read", "Write", "Execute"] });
15     }, 1000);
16   });
17 }
18
19 function generateSessionToken(profile) {
20   return new Promise((resolve) => {
21     setTimeout(() => {
22       console.log("Task 3 Completed: Session token signed for security level: ${profile.role}.");
23       resolve("JWT_SESSION_TOKEN_TOKEN_SECURE_AUTH_${profile.id}");
24     }, 2000);
25   });
26 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D:\MERN\48_Mansi_701_A1>cd prob12_chained_promises
D:\MERN\48_Mansi_701_A1\prob12 chained promises>node chain.js
Beginning chained sequence runtime execution...
Task 1 completed: User credentials validated.
Task 2 Completed: Permissions retrieved for Mansi Patel.
Task 3 Completed: Session token signed for security level: Administrator.
Final Response Output -> Active Session Token: JWT_SESSION_TOKEN_TOKEN_SECURE_AUTH_701
```

D:\MERN\48_Mansi_701_A1\prob12 chained promises